

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

INTERNATIONAL GCSE COMPUTER SCIENCE

Paper 2 Concepts and principles of computer science

Thursday 6 November 2025

07:00 UK Time

Time allowed: 2 hours

Materials

You will need no other materials.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of the page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this booklet. Cross through any work you do not want to be marked.
- You are not allowed to use a calculator.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.
- You may use a bilingual dictionary.
- You may **not** use an English dictionary.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
TOTAL	



N 0 V 2 5 9 2 1 0 2 0 1

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9210/2

Answer **all** questions in the spaces provided.

0 1 . 1 Convert the decimal number 35 to binary.

[1 mark]

Answer _____

0 1 . 2 Convert the decimal number 172 to hexadecimal.

[1 mark]

Answer _____

0 1 . 3 State **two** reasons why hexadecimal is often used by programmers instead of binary.

[2 marks]

Reason 1 _____

Reason 2 _____



0 1 . 4 Tick (✓) **two** boxes to show which of the statements are true.

[2 marks]

	Statement	Tick (✓) two
A	There are 4 bits in a byte.	
B	1 MB is greater than 200 000 bytes.	
C	1000 TB is equal to 1 GB.	
D	1 byte can be used to represent 256 different bit patterns.	
E	1 kB is equal to 1000 bits.	
F	8 bits can represent twice as many bit patterns as 4 bits.	

0 1 . 5 Describe how a binary shift can be used to multiply a binary number by 8

[2 marks]

8

Turn over for the next question

Turn over ►



0 2 . 1 Complete the truth table for the AND logic gate.

[1 mark]

X	Y	Q
0	0	
0	1	
1	0	
1	1	

0 2 . 2 Draw a logic circuit in the space below to represent how the heating in a house is controlled.

The heating (**H**) should be on when one or more of the following conditions are true:

- the timer (**T**) is on and the temperature sensor (**S**) has a value of true
- the button (**B**) is on.

[3 marks]



0	2	.	3
---	---	---	---

Write a Boolean expression that represents the circuit that is used to turn a light on or off.

The light (**L**) should be on when one or more of the following conditions are true:

- the door sensor (**D**) is on
- the pressure sensor (**P**) is off and the control switch (**C**) is off.

[3 marks]

L = _____

7

Turn over for the next question

Turn over ►



0 3

The control unit and clock are components of a CPU.

0 3 . 1

State the name of **one** other component of a CPU.

[1 mark]

0 3 . 2

Describe **one** task carried out by the control unit.

[1 mark]

0 3 . 3

Describe the purpose of a bus.

[1 mark]

0 3 . 4

The CPU processes instructions. The first stage in processing an instruction is the fetch stage. In the fetch stage an instruction is fetched to the CPU from the main memory.

State the names of the next **two** stages.

[2 marks]

Second stage _____

Third stage _____



0 3 . 5 Main memory consists of ROM and RAM.

State **two** differences between ROM and RAM.

[2 marks]

Difference 1 _____

Difference 2 _____

0 3 . 6 Explain why secondary storage is needed.

[1 mark]

0 3 . 7 Tick (✓) **one** box in each row to show the type of storage the statement is true for.

[4 marks]

	Statement	Solid state	Magnetic	Optical
A	Has no moving parts.			
B	Uses electrical circuits to store data.			
C	Has a spiral track.			
D	Uses a laser.			
E	Data is represented by microscopic pits.			

12

Turn over for the next question

Turn over ►



0 4**Figure 1** shows the HTML for a web page.**Figure 1**

```

<html>
  <head>
    <title>The world</title>
  </head>
  <body>
    <h1>Continents</h1>
    <h2>Africa</h2>
    <p>Africa facts:</p>
    <ul>
      <li>2nd largest continent</li>
      <li>Over 3000 languages</li>
    </ul>
  </body>
</html>

```

0 4 . 1Draw what the web page in **Figure 1** will look like when displayed in a web browser.

You may use labels to make your layout clearer, if necessary.

[4 marks]


0 4 . 2

The student wants to change the web page so that all the text is:

- in bold
- blue.

Complete the CSS below to make these changes.

[2 marks]

```
body {  
  
    font-_____: bold;  
  
    _____: blue;  
  
}
```

0 4 . 3

A user enters a password to access a secure area on a website.

Describe **one** example of a check on the password that could be made using a client-side scripting language.

[1 mark]

7

Turn over for the next question

Turn over ►



0 5

Linear search and binary search are two algorithms that can be used to find an item in a list.

0 5 . 1

Describe how a linear search works.

[3 marks]

0 5 . 2

A list contains nine items.

State the index of the first item that a binary search algorithm would compare to the item being searched for.

You should assume that the index of the first item in the list is 0

[1 mark]

0 5 . 3

A binary search algorithm is used on a list containing four items.

State the maximum number of items in the list that would be compared to the item being searched for.

[1 mark]

0 5 . 4

State **one** advantage of a binary search compared to a linear search.

[1 mark]



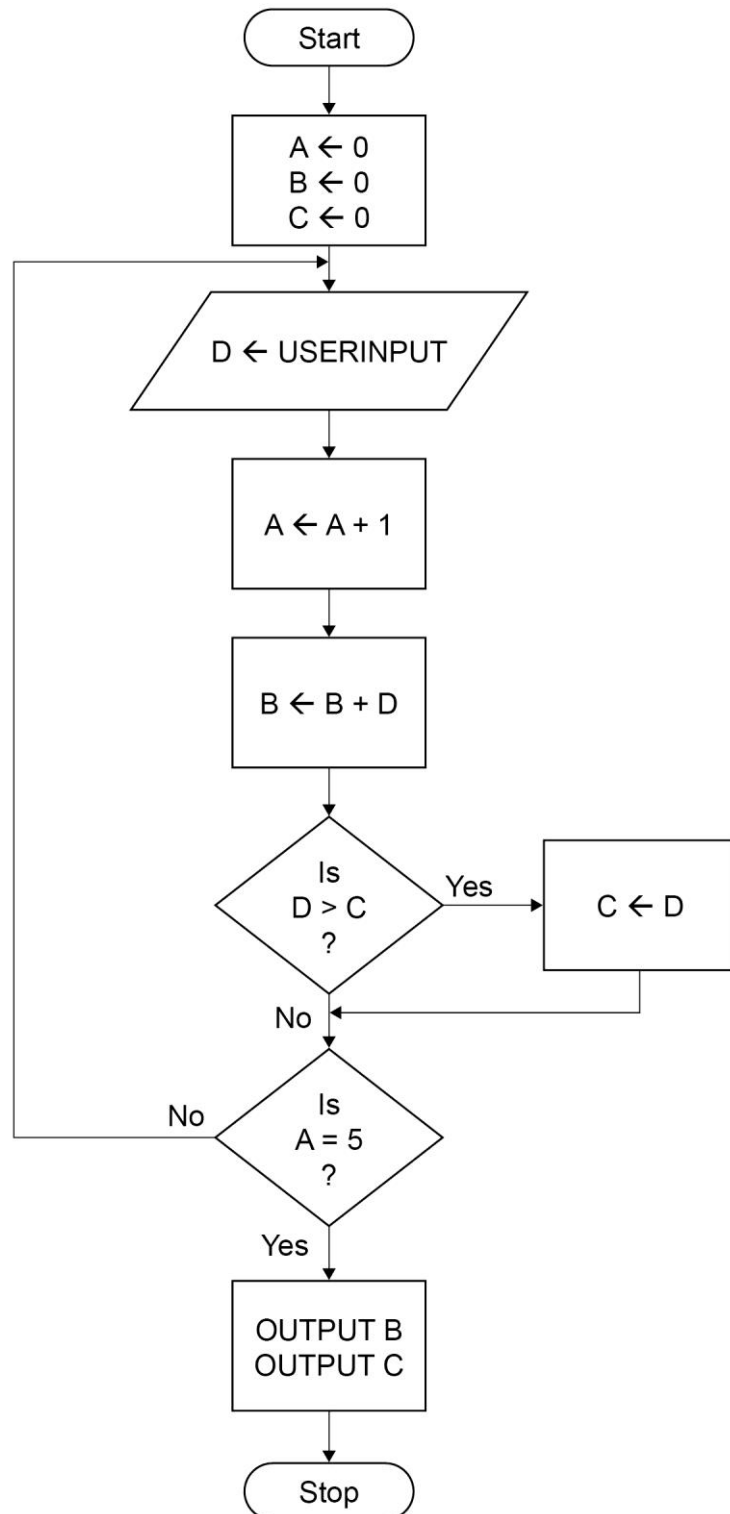
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ANSWER IN THE SPACES PROVIDED**

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0 6**Figure 2** shows a flowchart.**Figure 2**

0 6 . 1

Complete the trace table for the flowchart in **Figure 2** when the inputs are the five numbers:

5 6 3 8 2

You may not need to use all the rows.

[4 marks]

A	B	C	D

0 6 . 2

Describe what the values in the variables **B** and **C** represent when the algorithm represented by the flowchart in **Figure 2** stops.

[1 mark]

Variable **B** _____

Variable **C** _____

5

Turn over for the next question

Turn over ►



07.1

Define the term computer network.

[1 mark]

07.2

State **two** differences between a Wide Area Network (WAN) and a Local Area Network (LAN).**[2 marks]**

Difference 1

Difference 2

07.3

Describe the benefits of setting up a network as a wireless network instead of as a wired network.

[3 marks]



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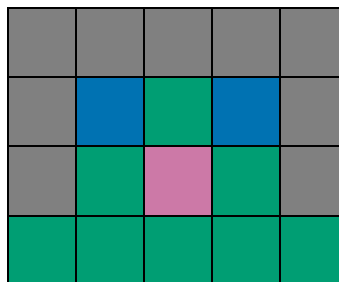
0 8 . 1 Bitmap images are made of pixels.

Describe what a pixel is.

[1 mark]

0 8 . 2 Figure 3 shows an image containing four colours.

Figure 3



Represent the image as a bitmap in binary.

The top row of pixels has already been represented.

[2 marks]

0 0 0 0 0 0 0 0 0 0

0	8	.	3
---	---	---	---

An image containing 52 colours is 5000 pixels wide and 800 pixels high.

Calculate the minimum file size for the image.

Give your answer in megabytes.

Show your working.

[3 marks]

Answer _____ megabytes

6

Turn over for the next question

Turn over ►



0 9 . 1

Describe what cyber security is.

[2 marks]

0 9 . 2

Explain why organisations use penetration testing.

[2 marks]

0 9 . 3

Describe what malware is.

[1 mark]

0 9 . 4State the name of **one** type of malware.**[1 mark]**

6

1 0 . 1

Describe how sound is sampled so that it can be stored digitally on a computer.

[3 marks]

1 0 . 2

A recording is made of a sound. The recording:

- lasts for 10 seconds
- uses a sample rate of 15 000 Hz
- uses a 16-bit sample resolution.

Calculate the amount of memory that will be needed to store the recording.

Give your answer in bytes.

Show your working.

[2 marks]

Answer _____ bytes

5

Turn over for the next question**Turn over ►**

1 1 . 1

The string `sleeveless` was compressed using Huffman coding.

The compressed string uses 19 bits of storage.

Calculate how many bits have been saved by representing the string using Huffman coding instead of ASCII.

Show your working.

[2 marks]

Answer _____ bits

1 1 . 2

Unicode can represent a greater range of characters than ASCII.

State **two** advantages of being able to represent a greater range of characters.

[2 marks]

Advantage 1 _____

Advantage 2 _____

4



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1 2

A database has been created to store data about the films that can be watched at a cinema. The cinema shows films on six different screens.

There are only two tables in the database: **Film** and **Showing**.

Figure 4 shows **some of the data** in the two tables.

Figure 4

Film

FilmID	FilmName	Category	LengthInMins
31	The Shadow Circuit	Action	114
32	Sheep Can't Drive	Comedy	93
33	The Clockmaker's Carnival	Animation	93
34	Infinity Breach	Action	135
35	Whispwood Tales	Animation	100
36	Rhythm Street	Musical	128

Showing

ShowingID	FilmID	ScreenNumber	Date	Time
401	31	4	09/03/2026	17:00
402	33	2	09/03/2026	20:30
403	31	1	10/03/2026	17:00
404	32	5	10/03/2026	17:10
405	36	1	10/03/2026	20:30

1 2

. 1

Tick (✓) **one** box to show which of the statements is false.

[1 mark]

	Statement	Tick (✓) one
A	A record is usually shown as a row in a table.	
B	A relational database can store data in more than one table.	
C	A table can only have one field in it that is a foreign key.	
D	Data redundancy is when the same data item is unnecessarily stored more than once.	
E	The primary key field in a table stores a unique value for each record.	

1 2

. 2

State the name of the field in the Showing table that is the foreign key.

[1 mark]



1 2 . 3

Write an SQL query that:

- finds the name, the time and the screen number
- of the Action or Comedy films
- showing on 10/03/2026

[5 marks]

1 2 . 4

The showing with ShowingID 401 has been cancelled. The record for this showing must be deleted from the database.

Write the first line of the SQL code to delete the record.

[1 mark]

Line 1 _____

Line 2 WHERE ShowingID = 401

8

END OF QUESTIONS

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